PARTIAL CHARACTER SUMS AND THE COLLISION TRANSFORM

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Abstract. For any integer $n > b$ and Dirichlet character χ modulo n, the collision transform satisfies the identity

$$\hat{C}_n(\chi) = \frac{1}{\phi(n)} \sum_{d=0}^{b-1} |A_d(\chi)|^2,$$

where $A_d(\chi) = \sum_{\substack{r \in B_d \\ \gcd(r,n)=1}} \chi(r)$ is the partial character sum over the d-th bin of the digit function $\lfloor br/n \rfloor$.

The collision invariant is the L^2 norm of the digit function's character spectrum resolved by bins. The components A_d are partial character sums over intervals, the central objects of analytic number theory. This identity connects the collision program to the classical theory of Burgess and Pólya–Vinogradov.

For composite $n = pq$, each A_d is a bilinear character sum over the CRT grid restricted to bin d. Computation at semiprimes up to $n = 667$ shows that ghost collisions (collisions present at the composite but absent at each prime individually) account for nearly all collisions. In the semiprimes tested, the bin matrices exhibit structured singular values encoding the interleaving of the two primes' digit geometries.

1. Introduction

Write $U(n) = (\mathbb{Z}/n\mathbb{Z})^*$ for the group of units. The collision count at multiplier $g \in U(n)$ is $C(g) = \#\{r \in U(n) : \lfloor br/n \rfloor = \lfloor b(gr \bmod n)/n \rfloor\}$, and the collision deviation is $S(g) = C(g) - \lfloor (n-1)/b \rfloor$. These were introduced in [**?**]. The collision transform [**?**] decomposes S over Dirichlet characters, and the decomposition theorem [**?**] gives an exact factorization encoding L-function special values.

These results operate at a fixed modulus $m = b^2$. The present paper works at the modulus n itself, where the Fourier transform is over $(\mathbb{Z}/n\mathbb{Z})^*$ and the characters, for $n = pq$, factor as $\chi = \chi_p \times \chi_q$.

The main result is that the collision transform decomposes into squared partial character sums over the digit function's bins. The

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identity holds at every modulus. It connects the collision program to the classical theory of partial character sums: the objects that Pólya, Vinogradov, and Burgess spent a century studying are the components of the collision spectrum.

2. The Two Transforms

Two transforms. Two groups. Two kinds of information.

Definition 1. *Transform 1* (universal): the Fourier transform of S viewed as a function on $(\mathbb{Z}/b^2\mathbb{Z})^*$, with characters modulo b^2. For primes, finite determination gives dependence only on $n \bmod b^2$. The decomposition theorem [**?**] factorizes it into Bernoulli numbers and diagonal character sums.

Transform 2 (internal): the Fourier transform of the collision count $C(g)$ over $(\mathbb{Z}/n\mathbb{Z})^*$, with characters modulo n.

Transform 1 encodes L-function special values. It does not see the factorization: it depends on $n \bmod b^2$, which is the same n, not its factors. Transform 2 sees the factorization. Computing a single $C(g)$ costs $O(n)$; computing all $A_d(\chi)$ costs $O(n\phi(n))$.

3. Finite Determination at Composites

Observation 2. For primes p, the Transform 1 collision deviation $S_\ell(p)$ depends only on $p \bmod b^{\ell+1}$ (finite determination, [**?**]). Computation suggests the same holds at composites: at $n = 77$ and $n = 91$ in base 10, the lag-1 deviation matches all primes in the same class modulo 100 through $p = 1291$.

Transform 2 (the full function $g \mapsto C(g)$ on $U(n)$) depends on n itself, not just on a residue class. It is Transform 2 that the bin character sum identity addresses.

4. The Bin Character Sum Identity

Definition 3. For $n > b$ and $0 \le d \le b-1$, the d-th bin is $B_d = \{r \in \{1, \ldots, n-1\} : \lfloor br/n \rfloor = d\}$. The *bin character sum* is $A_d(\chi) = \sum_{\substack{r \in B_d \\ \gcd(r,n)=1}} \chi(r)$.

Theorem 4 (Bin character sum identity)**.** *For any $n > b$ and any Dirichlet character χ modulo n:*

$$\hat{C}_n(\chi) = \frac{1}{\phi(n)} \sum_{d=0}^{b-1} |A_d(\chi)|^2,$$

where $\hat{C}_n(\chi) = \frac{1}{\phi(n)} \sum_{\substack{g=1 \\ \gcd(g,n)=1}}^{n-1} C(g)\,\overline{\chi}(g)$.

Proof. Expand:

$$\phi(n)\,\hat{C}_n(\chi) = \sum_g \sum_r \mathbf{1}[\lfloor br/n \rfloor = \lfloor b(gr \bmod n)/n \rfloor]\,\overline{\chi}(g),$$

where r and g run over units modulo n. Substitute $s = gr \bmod n$, so $g = sr^{-1}$ and $\overline{\chi}(g) = \overline{\chi}(s)\,\chi(r)$. The collision condition becomes $s \in B_{d(r)} \cap U(n)$:

$$= \sum_{r \in U(n)} \chi(r) \sum_{s \in B_{d(r)} \cap U(n)} \overline{\chi}(s).$$

Grouping by bin $d = d(r)$:

$$= \sum_{d=0}^{b-1} A_d(\chi)\,\overline{A_d(\chi)} = \sum_d |A_d(\chi)|^2. \qquad \square$$

The identity holds at every modulus, with no assumption on n (prime, composite, or prime power) and no restriction on χ (primitive, imprimitive, even, or odd). It is a substitution identity: the change of variables $g = sr^{-1}$ decomposes the collision kernel into bin-level components.

5. Connection to Partial Character Sums

The bin character sums $A_d(\chi)$ are partial sums of χ over intervals of length $H \approx n/b$. These are the objects studied by Pólya–Vinogradov [**?**] and Burgess [**?**]. Pólya–Vinogradov gives

$$|A_d(\chi)| \le C\sqrt{n}\,\log n$$

for non-principal primitive χ. Since each $A_d(\chi)$ is an incomplete character sum over an interval, Burgess bounds also apply and become relevant for shorter interval refinements.

Applying Pólya–Vinogradov to the identity:

$$|\hat{C}_n(\chi)| \le \frac{b}{\phi(n)}\,C^2 n\,\log^2 n \qquad \text{for non-principal primitive } \chi.$$

Conversely, the collision data determines $\sum_d |A_d|^2$ exactly. Advances in the evaluation of partial character sums over intervals would immediately yield sharper information about the collision spectrum.

Remark 5. The decomposition theorem [**?**] expresses Transform 1 as a product $B_1 \cdot \overline{S_G}$. The bin character sum identity expresses Transform 2 as a sum of squares $\sum |A_d|^2$. The two identities are complementary: one factors, the other decomposes.

6. Ghost Collisions

Definition 6. A *ghost collision* at multiplier g and residue r for $n = pq$ is a collision at the composite ($\lfloor br/n \rfloor = \lfloor b(gr \bmod n)/n \rfloor$) that is not a collision at either prime individually.

Observation 7. For the semiprimes tested in base 10 (Table **??**), ghost collisions account for 98–100% of all collisions. At $n = 7 \times 11 = 77$ (both primes digit-partitioning), the ghost fraction is exactly 100%.

n	factorization	ghost %	$\phi(n)$
77	7×11	100.0	60
91	7×13	100.0	72
143	11×13	100.0	120
221	13×17	99.4	192
391	17×23	98.2	352
667	23×29	98.7	616

Table 1. Ghost collision fraction at semiprimes in base 10.

The mechanism: the composite digit $\lfloor br/n \rfloor$ depends on the residue pair $(r_p, r_q) = (r \bmod p,\ r \bmod q)$, not on the individual prime digits $\lfloor br_p/p \rfloor$ and $\lfloor br_q/q \rfloor$ alone. Two residues with the same prime-level digits can have different composite digits. Ghost collisions arise when multiplication by g changes both prime-level digits but the composite digit stays the same, because the composite bin is wide enough (width pq/b) to absorb the shift that the prime-level bins (width p/b and q/b) reject.

7. The Interleaving and Its Rank

Definition 8. The *bin matrix* M_d for $n = pq$ is the $(p-1) \times (q-1)$ binary matrix indexed by $(i, j) \in (\mathbb{Z}/p\mathbb{Z})^* \times (\mathbb{Z}/q\mathbb{Z})^*$, with $M_d[i][j] = 1$ if $\mathrm{CRT}(i, j) \in B_d$.

Write $\boldsymbol{\chi}_p = (\chi_p(1), \chi_p(2), \ldots, \chi_p(p-1))^\top$ for the vector of character values on the units modulo p, and similarly for $\boldsymbol{\chi}_q$. The bin character sum factors as a bilinear form:

$$A_d(\chi_p \times \chi_q) = \boldsymbol{\chi}_p^\top M_d \, \boldsymbol{\chi}_q.$$

Observation 9. At $n = 91 = 7 \times 13$ in base 10: each bin matrix M_d has rank 5–6 (out of a maximum of $6 = p-1$). The singular values are all 1 or $\sqrt{2}$. The matrices are far from rank 1.

At $n = 91$, the high rank indicates that A_d does not approximately factor as $f(\chi_p) \cdot g(\chi_q)$. The tensor structure of the character group does not pass through the bin decomposition in this case.

8. The Ghost Correlation Void

The collision correlation function exhibits dips at factor-aligned distances.

Definition 10. For $d \in U(n)$, define the collision correlation

$$R(d) = \frac{\sum_{g \in U(n)} C(g)\, C(gd \bmod n)}{\sum_{g \in U(n)} C(g)^2}.$$

For a candidate modulus m, the *correlation excess* is

$$\varepsilon(m) = \overline{R}_{d \equiv 1} - \overline{R}_{d \not\equiv 0,1},$$

where the bars denote averages over $d \in U(n)$ satisfying the indicated congruence condition modulo m.

Observation 11. For the composites tested (Table **??**): when m is a factor of n, $\varepsilon(m) < 0$; when m is not a factor, $\varepsilon(m) \approx 0$.

n	factors	excess (factors)	excess (non-factors)
667	$23, 29$	$-0.031,\ -0.036$	± 0.003
1147	$31, 37$	$-0.023,\ -0.029$	± 0.002
1763	$41, 43$	$-0.020,\ -0.022$	± 0.002

Table 2. The ghost correlation void: factors produce negative excess, non-factors do not.

The excess scales approximately as $\varepsilon \sim -c/p$. Detection requires $O(1/\varepsilon^2) = O(p^2) = O(n)$ samples, which does not improve on direct computation. The void is a structural signature of the factorization, not a computational shortcut.

9. The Factoring Wall

The factorization of n is encoded in the bin character sums $A_d(\chi_p \times \chi_q)$. The wall is not in the encoding. It is in the reading.

Observation 12. Direct computation of $A_d(\chi)$ for a single bin and character costs $O(n/b)$ (summing over all residues in the bin). The total cost for all bins and characters is $O(n \cdot \phi(n))$.

For primitive χ, the Gauss sum expansion

$$A_d = \tau(\overline{\chi})^{-1} \sum_a \overline{\chi}(a) \sum_{r \in B_d} e(ar/n)$$

reduces the inner sum to a geometric series with explicit evaluation, but the outer sum has $\phi(n)$ terms.

Within the representations considered here, the cost remains proportional to the grid's size. The digit function projects a two-dimensional structure onto a one-dimensional assignment. The A_d are the projections. They carry the factorization in their phases, and their squared magnitudes in the collision transform.

10. Remarks

The gate width at composites. The gate width theorem [**?**] states that exactly $b-1$ multipliers are deranging ($C(g) = 0$) for any prime $n > b$. At composites, the count can exceed $b-1$. At $n = 91$ in base 10: 12 deranging multipliers instead of 9. The 3 extras satisfy $g \equiv 1 \pmod{13}$ and reveal the factor via $\gcd(g-1, n) = 13$.

Composite antisymmetry. For non-principal $\chi_p \times \chi_q$ with $\chi_p(-1) \cdot \chi_q(-1) = +1$ (both characters of the same parity), $B_1(\chi_p \times \chi_q) = 0$. Non-trivial even composite characters have vanishing Bernoulli numbers.

The tower. The decomposition theorem at lag ℓ gives L-value data at characters modulo $b^{\ell+1}$. The tower of data across lags $\ell = 1, 2, 3, \ldots$ suggests a p-adic structure whose growth rate may encode Euler factors of the L-functions involved. The tower ratios differ between composites and primes (verified at $n = 91$ vs $p = 191$ in base 5). The relationship between the tower and the bin character sum identity is under investigation.

Open directions. The bin character sum identity reduces the collision transform to partial character sums over intervals. Advances in evaluating such sums (beyond the Burgess bound) would directly sharpen the collision spectrum at composites. The bilinear structure of the bin matrices, with their restricted singular values, may admit a theoretical characterization in terms of the CRT geometry.

The p-adic tower of decomposition theorems may connect the collision program to Iwasawa theory. The tower ratios at different lags give L-value data at increasing resolution. Whether this data constrains the factorization beyond what $n \bmod b^{\ell+1}$ provides at each individual level is the central open question of the program.

References

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